

BUIDL_OPC HACKATHON · SINGAPORE

FarmOS

The operating system for a one-person farm company.

Track: OPS / Super Individuals

Built with Claude + AWS

100-acre eco-farm · Malaysia

THE PROBLEM

One person. Ten businesses. A hundred off-grid acres.

A single founder is opening lodging, a restaurant, hydroponics, aquaponics, an orchard, poultry, a petting zoo, palm oil, recycling and bees — on one 100-acre site in Malaysia, with its own solar and water and no reliable grid.

Each vertical is a full business with its own sensors, staff and failure modes. Run by hand, this needs a team of ten. **One person can't hold it in their head — and if the internet drops, the tools stop.**

THE INSIGHT

Not ten businesses. One platform, ten modules.

Treat the farm as a single operating system with plug-in modules on one shared life-support layer. That's what makes it runnable by one person — and replicable to the next farm from a config file.

THE SOLUTION

FarmOS: a digital twin + an autonomous Claude agent

The agent is the operations team. It reads the whole farm, decides across every vertical, acts through real devices, and logs every decision — so one founder runs ten departments.

Autonomous

A Claude supervisor agent perceives, decides and acts across ten domains with no human in the loop.

Sovereign

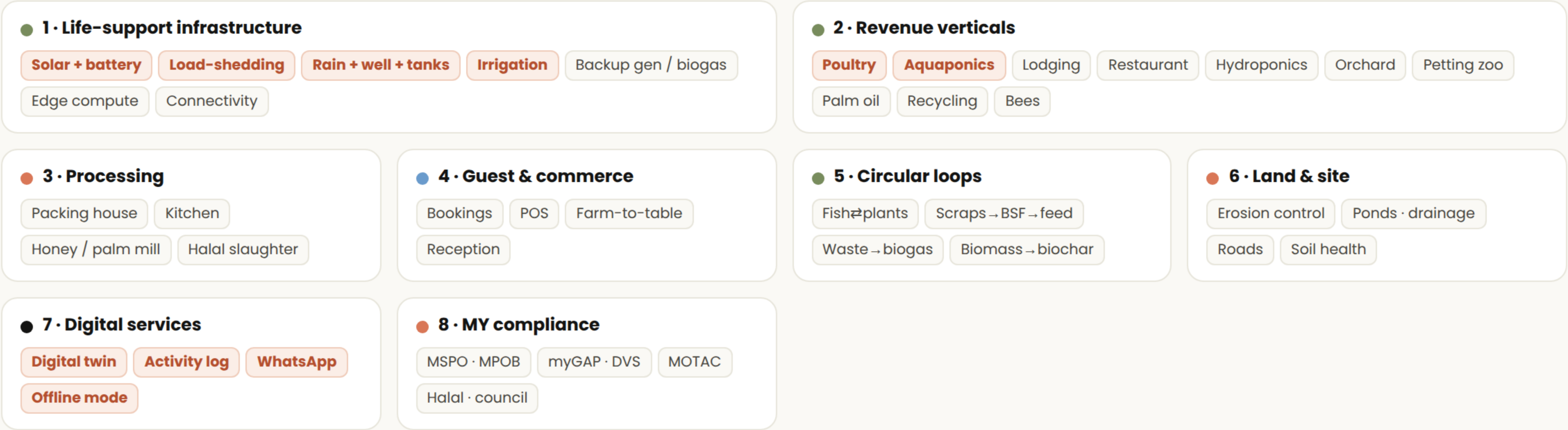
Runs on-site on solar and local compute.
Keeps operating through a grid + internet blackout.

Replicable

A new farm is a new config file, not a new build. One system, many sites.

WHAT IT RUNS — THE PLATFORM MAP

Eight system layers · one agent · ten verticals



HOW IT WORKS — REAL HARDWARE

Off-the-shelf devices → one on-site hub → the agent



Cut the cord: the hub runs on-site on solar — it keeps working with no internet. **Demo = farm:** flip simulated: false and the same agent moves real hardware.

THE AUTONOMY

“What my company did while I slept.”

02:14 — Battery 22%, cloudy dawn forecast. Shed the pool pump & EV charger; protected the incubator, aquaponics aerator and cold room.

05:40 — Rain forecast in 6h. Skipped orchard irrigation, saved 4,000L.

06:02 — Coop 3 hit 34°C. Fans on, WhatsApp'd the owner.

The agent's tools

getFarmState()

getForecast()

shedLoad()

setActuator()

scheduleIrrigation()

createAlert()

assignTask()

Every call is logged with the agent's reasoning — the founder wakes up to a full account of the night, and trusts the system more each day.

SOVEREIGNTY

When the world goes dark, the company keeps running.

Solar and rainwater for power and water. Home Assistant and the agent's fallback logic run on a Raspberry Pi on-site. In a Malaysian monsoon blackout, FarmOS sheds non-critical loads, keeps life-support alive, and queues the outside world for when it returns.

Energy sovereignty

Water sovereignty

Food sovereignty

Digital sovereignty

IT'S REAL, NOT A MOCKUP

Every asset maps to a product you can buy today

Sense

Atlas Scientific pH/EC/DO · SenseCAP & Dragino
LoRaWAN soil · Ecowitt weather · BroodMinder hives

Automate

Shelly & Sonoff relays · OpenSprinkler irrigation ·
ChickenGuard coop door · Atlas dosing pumps

Robots

FarmBot (hydroponics) · Husqvarna Automower
(orchard) · automatic feeders · drone imagery

Energy & water

Victron Cerbo GX · Solar Assistant · ultrasonic tank
levels · solar borehole pump

Hub & brain

Home Assistant + Node-RED + MQTT on a Raspberry
Pi · Claude agent on AWS

Business

Beds24 bookings · Loyverse POS · Twilio WhatsApp ·
Wave / Odoo accounting

Live demo flow

- 1 **Live farm dashboard** — the whole company as real-time data.
- 2 **“Overnight” agent log** — every autonomous decision and its reasoning.
- 3 **Crisis** — low battery + cloud → the agent sheds loads and protects life-support, live.
- 4 **Cut the cord** — kill grid + internet; FarmOS keeps running autonomously.
- 5 **Replicate** — boot “Farm #2” from one config file.

Roadmap

TODAY

The MVP

Digital twin + autonomous agent + the two demo scenarios. Fully simulated — nothing to buy.

PHASE 1

Prove one loop

Raspberry Pi + Home Assistant, a few Shelly relays and an Atlas probe. Aquaponics + poultry go live.

PHASE 2

Scale the farm

LoRaWAN sensor fleet, irrigation, energy monitoring, bookings + POS. The full 100 acres online.

PHASE 3

Franchise-in-a-box

Robots + a config template. Replicate FarmOS to Farm #2, #3 — a thousand sovereign farms.

One person. Ten businesses. One operating system.

Runs through a blackout. Replicable to the next farm.

Built with Claude + AWS

FarmOS · Malaysia